§6 Capacitors & Inductors

(def) $q = Cv$

$i = C\frac{dv}{dt}$

$v = \frac{1}{C}\int_{t_0}^{t} i\,dt + v(t_0)$

$w = \frac{1}{2}Cv^2$

series $C_{eq} = \sum C_i$

parallel $\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$

(def) $\psi = Li$

$v = L\frac{di}{dt}$

$i = \frac{1}{L}\int_{t_0}^{t} v\,dt + i(t_0)$

$w = \frac{1}{2}Li^2$

series $L_{eq} = \sum L_i$

parallel $\frac{1}{L_{eq}} = \sum \frac{1}{L_i}$

§7 First-Order Circuits

RC Circuits

$\tau = RC$

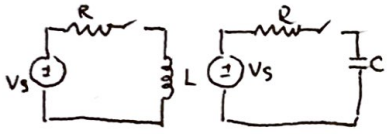
$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-\frac{t}{\tau}}$

$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}}$

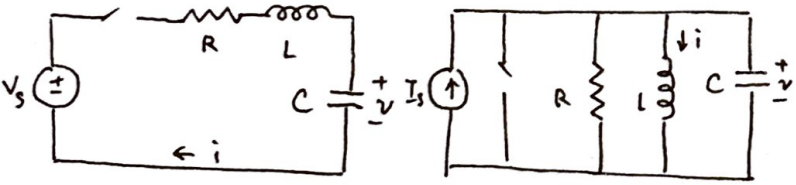
RL Circuits

$\tau = \frac{L}{R}$ (equivalent resistence at L)

$*\ e^{-\frac{5\tau}{5}} \approx 0$



§8 Second-Order Circuits



Series RLC Circuit

$\begin{cases} iR + L\frac{di}{dt} + v = 0\ (V_s) \\ i = C\frac{dv}{dt} \end{cases}$

$\Rightarrow s^2 + \frac{R}{L} + \frac{1}{LC} = 0\ (\frac{V_s}{LC})$

$\alpha = \frac{R}{2L}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{1}{LC}v = \frac{1}{LC}v_c$

Parallel RLC Circuit

$\begin{cases} \frac{v}{R} + i + C\frac{dv}{dt} = 0\ (I_s) \\ v = L\frac{di}{dt} \end{cases}$

$\Rightarrow s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0\ (\frac{I_s}{LC})$

$\alpha = \frac{1}{2RC}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{1}{LC}i = \frac{1}{LC}I_s$

① Over-damping $\alpha > \omega_0$

$x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

② Critical damping $\alpha = \omega_0$

$x(t) = (B_1 + B_2 t)e^{st}$

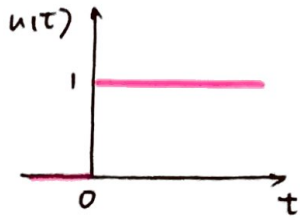
③ Under-damping $\alpha < \omega_0$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$x(t) = e^{-\alpha t}(C_1\cos(\omega_d t) + C_2\sin(\omega_d t))$

§7.4 Singularity Functions

① Unit Step

$u(t) = \begin{cases} 0, & t<0 \\ 1, & t>0 \end{cases}$

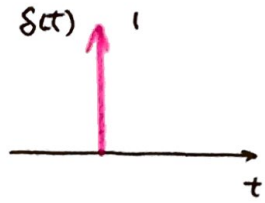


② Unit Impulse

$\delta(t) = \frac{d}{dt}u(t)$

$u(t) = \int_{-\infty}^{t}\delta(t)\,dt$

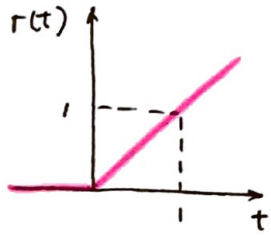
$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\,dt = f(t_0)$



③ Unit Ramp

$r(t) = \begin{cases} 0, & t \le 0 \\ t, & t \ge 0 \end{cases}$

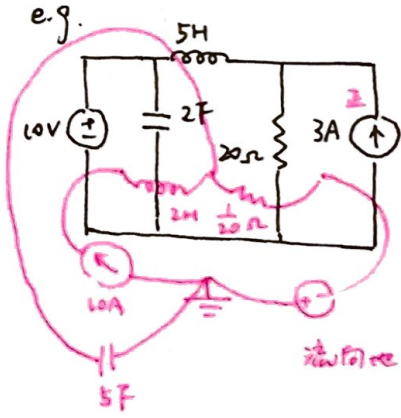
$r(t) = \int_{-\infty}^{t} u(\tau)\,d\tau = tu(t)$



§8.10 Duality

R	G
L	C
v	i
±	↑
Node	Mesh
Series	Parallel
Open	Short
KVL	KCL
Thevinin	Norton

e.g.



§11 AC Power Analysis

$p(t) = v(t)i(t) = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m\cos(2\omega t + \theta_v + \theta_i)$

$P = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) = \frac{1}{2}\mathrm{Re}(\tilde{V}\tilde{I}^*)$

$= \frac{1}{2}I^2R$

If $\theta_v = \theta_i$

$P = \frac{1}{2}V_m I_m = \frac{1}{2}I_m^2 R = \frac{V_m^2}{2R}$

If $\theta_v - \theta_i = \pm 90°$

$P = 0$

§9 Sinusoids & Phasors

sinusoid : $v(t) = V_m \cos(\omega t + \varphi)$

phasor : $\tilde{V} = V_m e^{j\varphi} = V_m \angle \varphi$

sinor : $\tilde{V} e^{j\omega t} = V_m e^{j(\omega t + \varphi)}$

$\Rightarrow v(t) = Re(\tilde{V} e^{j\omega t})$

Element	Time domain	Frequency domain
R	$v = Ri$	$\tilde{V} = R\tilde{I}$
L	$v = L\frac{di}{dt}$	$\tilde{V} = j\omega L \tilde{I}$
C	$i = C\frac{dv}{dt}$	$\tilde{V} = \frac{1}{j\omega C}\tilde{I}$

§11.3 Maximum Average Power Transfer

$Z_L = Z_{Th}^*$ $\quad P_{max} = \frac{V_{Th}^2}{8R_{Th}}$

§11.4 Effective / RMS Value

$P = I_{eff}^2 R \Rightarrow \begin{cases} I_{eff} = \sqrt{\frac{1}{T}\int_0^T i^2 dt} = I_{rms} \overset{\text{sinusoid}}{=} \frac{I}{\sqrt{2}} \\ V_{eff} = \sqrt{\frac{1}{T}\int_0^T v^2 dt} = V_{rms} = \frac{V}{\sqrt{2}} \end{cases}$

§10 Steady State Analysis

KCL KVL

Δ - Y conversion

Nodal Analysis

Mesh Analysis

Thevinin's Theorem

Norton's Theorem

Superposition

Source Transformation

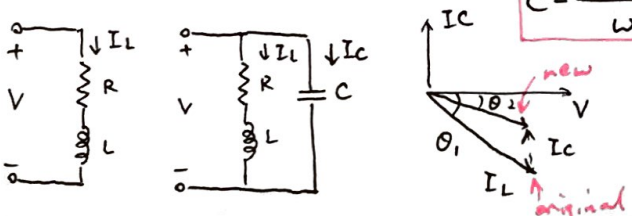
$S = \tilde{V}_{rms}\tilde{I}_{rms}^* = \frac{V_{rms}^2}{Z^*} = I_{rms}^2 Z$

$P = I_{rms}^2 R \quad Q = I_{rms}^2 X \quad (Z = R + jX)$

$= \frac{V_{rms}^2}{R} \quad = \frac{V_{rms}^2}{X}$

§11.8 Power Factor Correction

$$\boxed{C = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V^2}}$$



§9.5 Impedance & Admittance

$Z = \frac{\tilde{V}}{\tilde{I}}\ (\Omega) \qquad Y = \frac{\tilde{I}}{\tilde{V}}\ (S)$

$= R + jX \qquad = G + jB$

resistance reactance conductance susceptance

$X > 0$ inductive / lagging (i lags v)

$X < 0$ capacitive / leading (i leads v)

§11.5 Apparent Power & Power Factor

§11.6 Complex Power

$S = \tilde{V}_{rms}\tilde{I}_{rms} = |S| \angle(\theta_v - \theta_i)$

VA

$= V_{rms} I_{rms} \cos(\theta_v - \theta_i) + j V_{rms} I_{rms} \sin(\theta_v - \theta_i)$

$= P + jQ$

W VAR

power factor angle

$pf = \cos(\theta_v - \theta_i)$

① $Q = 0$ resistive load. unity pf

② $Q < 0$ capacitive ~ leading pf

③ $Q > 0$ inductive ~ lagging pf

* $e^{i\theta} = \cos\theta + i\sin\theta = \angle\theta$

$x + iy = \sqrt{x^2+y^2}\, e^{i\arctan\frac{y}{x}} = \sqrt{x^2+y^2}\angle\arctan\frac{y}{x}$

* $\frac{x\angle\alpha}{y\angle\beta} = \frac{x}{y}\angle(\alpha - \beta)$

$(x\angle\alpha)(y\angle\beta) = xy\angle(\alpha+\beta)$

* +, − rectangular form

×, ÷ angular form

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